A-A-1197A
February 4, 1987
SUPERSEDING
A-A-1197
June 9, 1981

COMMERCIAL ITEM DESCRIPTION

RUBBING COMPOUND, LACQUER

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers rubbing compound for use on lacquered surfaces to produce a smooth and high gloss finish.

Salient characteristics:

TYPE I - Shall be furnished in liquid form.
TYPE II - Shall be furnished in paste form.

Shall be a homogeneous mixture containing a finely divided abrasive.
Shall contain no material which would constitute a a health hazard when used as intended and must not produce obnoxious vapors in such a concentration as to constitute a health hazard or annoyance.
Shall contain no chlorinated hydrocarbons except that it may contain Dichloromethane and Trichlorethylene and Perchlorethylene of low concentrations.
Shall be furnished as a liquid or paste as specified.
Shall not chemically attack a lacquered surface when used as intended.
Shall not discolor or attack an aluminum clad aluminum alloy surface when used as intended.
Shall have a pH value of not more than 10.7.
Shall have a flash point not less than 140 F. (60 C).

Workmanship. The ingredients shall be homogeniously blended and processed as to produce a rubbing compound which is stable and not subject to abnormal change with age in a sealed container nor have any other defects affecting its serviceability.

FSC 5350

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Quality Assurance:

Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Sampling. Sampling and acceptance shall be in accordance with the applicable Methods 1022 and 1031 of Fed. Test Method Std. No. 141.

Test Procedures:

Flash Point. Flash point values shall be determined by testing as prescribed in ASTM D93.

Chemical attack. Vapor degrease two 3 inch x 6 inch panels of 20-gage cold rolled 1020 steel strip and spray with a coat of primer conforming to Federal Specification TT-P-664. Allow to air-dry for 4 hours. Spray 2 coats of olive drab automotive lacquer conforming to test lacquer in table I and allow to air-dry for 24 hours. Coat both panels with the rubbing compound in accordance with the manufactuer's instructions and allow to stand in room temperature 25 C. (77 F.) for 24 hours. Wipe one-half of each panel clean and examine the lacquer surface for any sign of chemical attack. Allow the panels to stand an additional 24 hours, and wipe the remaining portions clean. Examine the lacquer surface for any sign of chemical attack

Corrosiveness. Coat two 1 inch x 6 inch panels of an aluminum alloy 2024 core with thin layers of an aluminum alloy 1230 (99.3 % minimum aluminum) anodic to the core and of approximately equal thickness, bonded to both surfaces, with the rubbing compound in accordance with the manufacturer's instructions and allow to stand at room temperature 25 C. (77 F.) for 24 hours. Wipe one-half of each panel clean and examine the cleaned surfaces for corrosive attack, staining, or discoloration. Allow the panels to stand an additional 24 hours and wipe the remaining portions clean. Examine the cleaned surfaces for corrosive attack, staining, or discoloration.

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Rubbing efficiency.
Rubbing block. The rubbing surface of the block shall be 4 inches long by 3 inches wide and designed so that a strip of flannel may be securely clamped to and strecthed taut over the block in order to completely cover the bottom rubbing area. Provision shall be made for the attachment of lead weights to the top of the block in an evenly apportioned manner to give total weights of 8 and 15 pounds. The block shall be operated in a level, smooth, reciprocating motion through a length of 2 inches in a direction palallel to the length of the block. The block shall operate at a speed of 40 strokes per minute, a stroke being considered as one complete cycle from starting position of block and return to starting position.

Panels. Four steel panels 3 inches x 5 inches x 0.125 inch thickness of 1020 steel, one face of which shall be machined flat, shall have the flat face coated with one coat of primer,TT-P-664, and two coats of olive drab automotive lacquer conforming to test lacquer in Table I, so that the total dry film thickness is 1.5 to 1.7 mil. After air drying for 10 days the panels shall be washed vigorously with a mild soap solution, rinsed, and dried thoroughly and weighed to the nearest milligram.

TABLE I. Test lacquer

Ingredient	Percent by weight
Yellow dispersion (1)	8.0
Orange dispersion (2)	8.0
White dispersion (3)	2.0
Black dispersion (4)	6.0
Nitrocellulose RS 1/2 second (65 % in ethanol)	7.0
Non-drying phthalic alkyd resin (5) (65 % in xylol)	12.5
Blown castor oil (6)	1.6
Dioctyl phthalate	1.6
Ethyl acetate	6.6
Butyl acetate	13.3
Ethyl alcohol	4.3
Butyl alcohol	7.7
Toluol	21.4

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	Percent by weight
(1)Shall consist of the following:	
Chrome yellow lemon	50.0
RS 1/2 second nitrocellulose	11.0
Ethyl alcohol	4.5
Toluol	20.5
Ethyl acetate	14.0
(2)Shall consist of the following	
Chrome orange medium	60.0
RS 1/2 second nitrocellulose	17.0
Ethyl alcohol	7.0
Ethyl acetate	6.5
Toluol	9.5
(3)Shall consist of the following	
Titanium dioxide	60.0
RS 1/2 second nitrocellulose	8.0
Ethyl alcohol	3.5
Ethyl acetate	16.0
Toluol	12.5
(4)Shall consist of the following	
Carbon black	8.5
RS 1/2 second nitrocellulose	20.0
Ethyl alcohol	8.0
Ethyl acetate	25.0
Toluol	38.5

(5)Shall consist of the following

Non-drying phthalic resin:	
Phthalic anhydride	percent 35
Castor oil	percent 45

(6)Shall have the following characteristics	(1)
Viscosity (Gardner-Hold)	10-15
Acid number	50-60
Iodine number	201
Saponification number	11
Color (Gardner color standard)	

(1) heavier than Z-6

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Procedure. A weighted panel, prepared as specified above, shall be held securely in position beneath the rubbing block with the 5 inch dimension parallel to the length of the block.A fresh strip of plain weave,3.5 ounce,cotton flannel cloth (napped both sides) shall be clamped on the rubbing block. Weights shall be added to obtain a total rubbing block weight of eight pounds. Two ml. of the rubbing compound shall be measured by means of a calibrated plunger assembly, care being taken to eliminate air pocket. The rubbing compound shall be spread, using a spatula, as evenly as possible over the portion of the flannel which will contact the panel. The rubbing block shall be placed in position and moved through 100 strokes. Remove the rubbing block and discard the flannel. Wash the panel thoroughly using a mild soap solution and a soft bristle brush, dry thoroughly and weigh. Repeat the above procedure on a second panel, using fresh flannel and rubbing compound. Subject the third and fourth panels to the entire procedure listed above except that total rubbing block weight of 15 pounds and only 40 strokes shall be used. Fresh flannel and rubbing compound shall be used for each panel. The first and second panels tested above shall be paired as one set of panels, and the third and fourth panels shall be considered as the other set of panels.

Results. The loss in weight of each panel shall be calculated in milligrams. The average weight loss in each set of two panels shall be calculated. The sum of the two averages shall greater than 10 milligrams and less than 75 milligrams.

Appearance. Examine each panel after completion of the rubbing efficiency test. The lacquer surface shall be smooth, free from orange peel and scratches, and shall be sufficiently thick to completely hide the primer undercoat.

pH value. Place a 10 plus or minus 0.5 gram sample of the rubbing compound in a 100-ml. beaker and add 75 ml. of distilled water. Stir the mixture vigorously for one minute and determine the pH value electrometrically to the nearest 0.1 pH unit.

Preservation, packaging, packing, labeling, and marking (Civil Agencies. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Packaging (DOD). The packaging shall be as prescribed in ASTM D3951.

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Notes.
Purchaser shall specifiy TYPE I or TYPE II.

ASTM Standards are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

Federal Test Method Standard No. 141 is available from the Commander, Naval Publications and Forms Center, 5801 Taylor Avenue, Philadelphia, PA 19120.

The issues of those standards and specifications referenced in this document shall be those in effect on the date of the invitation for bids or request for quotation shall be used to determine compliance with these requirements.

MILITARY INTERESTS

Military coordinating activity:

Navy - YD

Review Activities:

Army - MR
Navy - SH, AS
Air Force - 99

User Activities:

Army - MI, ME, AT
Navy - MC

Preparing Activity:

GSA-FSS

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